OpenADMET CYP Challenge Report

Overview

We experimented with 3 main model architectures (Multi-task Molecular ICL, Graph Neural Networks, and classical Tabular Foundation Models). Placing well on the leaderboard required an ensemble of these models, as is common in these challenges. Since leaderboard signal was so high (50% of the total blind set), it was both possible and essential to understand how well cross validation (CV) splits tracked the leaderboard metrics. Agents helped curate data, prototype, and run+analyze hyperparameter tuning jobs within our workflow orchestration.

Datasets

We used the following public assays:

- OpenADMET CYP Challenge: Inhibition, TDI, Emax, single-concentration data
- PubChem CYP1A2: AID 410
- PubChem CYP2C9: AID 777, AID 883
- PubChem CYP2D6: AID 891
- PubChem CYP3A4: AID 884
- Veith et al. (2009): AID 1851
- Octant Bio / OpenADMET: CYP3A4 inhibition
- Kabir et al. (2022): CYP3A4/CYP3A7 inhibition

Models

Each of our model types was capable of individually placing in roughly the top 10 by spearman rank on the leaderboard (described below). Our final submission was an ensemble of these three model types, weighted based on individual performance and diversity.

1) Multi-task Molecular In Context Learning

We use an internal molecular ICL foundation model that supports multiple molecular featurization strategies and multitask learning across outputs. Our architecture incorporates structured task metadata into the ICL framework, allowing the model to leverage prior knowledge about different properties without requiring extensive historical data to be provided as context for each new task. Unlike Tabular Foundation Model pre-training, which is based on synthetic data generated from a Structural Causal Model, we instead design domain-specific priors informed by medicinal chemistry to improve performance.

2) Multi-task Graph Neural Networks

Because there are a number of heterogeneous data sources for CYP inhibition and the challenge set itself is fairly large and diverse, some of what gives ICL its edge in project-like settings is irrelevant. We trained a large number of GNNs starting from diverse architectures, but the two strongest contenders in the end were our internal architecture that alternates full attention with sparse message passing and CheMeleon finetuned in a similar way. We

pretrain our GNNs on a diverse set of tasks, one of which is the distillation task discussed here. We found it very helpful to have a student model trained to predict 3 of the 4 isoforms as a starting point.

3) Generic Tabular Foundation Models on OOF/Non-challenge Predictions

Diversity was hard to find when hyperparameter tuning models on the same fairly finite dataset, so we sought it in feature-space as well. We took models trained only on public, non-challenge datasets and labeled the train and test set, added SMARTS features for groups our other approaches seemed biased on, and threw it all into Kumo, TabICL, or TabFM. This was a strong single submission, but was made even stronger once out-of-fold (OOF) predictions for other models were allowed to be included as features, though we avoided stacking beyond this single level.

Splitting

We tried several splitting strategies and evaluated metrics on both the raw challenge pIC50s and those augmented by a learned transform from single-point to dose-response data. All splitting methods were able to differentiate strong and poor submissions, however none translated well to the top of the leaderboard.

Balanced splitting:

In this mode, we assign molecules to clusters, then clusters to folds. During this assignment, we stratify by assay values, attempting to avoid the scenario where all clusters containing potent compounds are in the same fold, say. We find this to report realistic metrics on ongoing campaigns and produce solid ensembles of models, but on some datasets it approximates (cluster --> random assignment).

Pseudo-Challenge splitting:

Here we took the top compounds in the training set, found their neighbors in the training set, and then split them unequally between two folds. That way, each fold would have some "dense" highly-active molecules chosen deterministically but still having analogs in other folds, but the rest would be assigned randomly.